

Creating identity matrix

```
>>> a = tf.eye(4)
>>> a.eval()
array([[ 1.,  0.,  0.,  0.],
       [ 0.,  1.,  0.,  0.],
       [ 0.,  0.,  1.,  0.],
       [ 0.,  0.,  0.,  1.]], dtype=float32)
```

Creating diagonal matrix

```
>>> r = tf.range(1, 5, 1)
>>> r.eval()
array([1, 2, 3, 4], dtype=int32)
>>> d = tf.diag(r)
>>> d.eval()
array([[1, 0, 0, 0],
       [0, 2, 0, 0],
       [0, 0, 3, 0],
       [0, 0, 0, 4]], dtype=int32)
```

Visualising data using TensorBoard

TensorBoard is a software utility that accepts the graphical representation of the data flow graph. The interpretations of the results are handled by the dashboard.

To launch TensorBoard, type `<url>:6006` into the Web browser. You can run it by using the command given below on a Windows computer:

```
<base> C:\Users\
Administrator>tensorboard-logdir-
summaries
TensorBoard 1.13.1 at http://SK-
DK-24:6006 (Press CTRL+C to quit)
```

Let's now attempt to build machine learning models with TensorFlow.

Linear and logistic regression with TensorFlow

Functions and differentiability are the basic concepts we need to know in order to understand machine learning. A function 'f' is a protocol that accepts an input to an output. A mathematical function is generally applied in basic physics and engineering. Differentiability is another type of control on functions. The fundamental advantage of the differentiable function

```
Administrator: Anaconda Prompt - pip install tensorflow
optional arguments:
  -h, --help            Show this help message and exit.
  -V, --version          Show the conda version number and exit.

conda commands available from other packages:
  build
  convert
  debug
  develop
  env
  index
  inspect
  metapackage
  render
  server
  skeleton
  verify

(base) C:\Users\Administrator>pip install tensorflow
Collecting tensorflow
  Downloading https://files.pythonhosted.org/packages/7b/14/e4538c2bc3ae9f4ce6f6ce7ef1180da05abc4a617afba798268232b01d0d/tensorflow-1.13.1-cp37-cp37m-win_amd64.whl (63.1MB)
    100% |#####| 63.1MB 193kB/s
```

Figure 3: Installation of TensorFlow

is to use the slope of the function at a corresponding position to find out the places where the function is higher or lower than the present position. This helps to find out the *minima* of the function. For instance, if the function is at a higher level, the machine learning problems are solved by the *loss* function, where the minima of the function is encoded with real world problems to provide solutions.

A regression function generally uses L2 loss where the magnitude of a vector is measured.

$$\|x\|_2 = \sqrt{\sum_{i=1}^N x_i^2}$$

Here, x is a vector of length N and L2 norm is generally defined as the distance between two vectors.

$$\|x-y\|_2 = \sqrt{\sum_{i=1}^N (x_i - y_i)^2}$$

Let's suppose that 'a' is a group of data, 'b' represents the corresponding labels, and 'f' is the differentiable function that encrypts the machine learning model. Since 'f' creates a prediction of 'b', then the L^2 loss function will be determined as:

$$L(a,b) = \|f(a)-b\|_2$$

Learning with TensorFlow

To install TensorFlow, use the command line:

```
C:\Users\Administrator>pip install
tensorflow
```

Use the command given below to install Jupyter Notebook:

```
<base> C:\Users\Administrator>pip
install jupyter
```

The TensorFlow library works under the alias of 'tf' and initialises

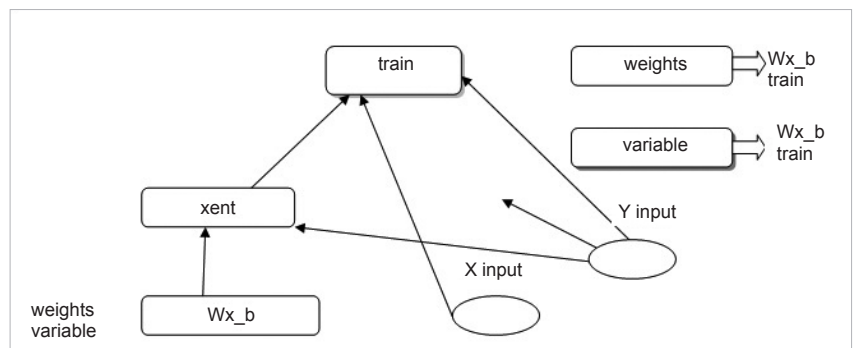


Figure 4: Data flow diagram for TensorBoard